CO-REGISTRATION ANALYSIS OF COHERENT MULTITEMPORAL RADARGRAMS

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***Abstract*—Radar Sounder data provide insights into subsurface geology and are typically analyzed using automatic techniques on single-pass acquisitions. A few preliminary studies have explored multitemporal RS data from the cryosphere. However, challenges related to data co-registration exist because of the acquisition geometry and the characteristics of the targets. This paper examines co-registration accuracy in overlapping bi-temporal RS acquisitions co-registered with coherent and non-coherent approaches to evaluate the importance of the phase signal. We project and co-register overlapping radargram pairs to mitigate the horizontal changes related to the ice motion and vertical variations in mass balance. We used the log ratio to compare the co-registered data and highlight backscattering differences in the radargrams due to residual misregistration (and morphological changes). We analyze overlapping radargram pairs acquired in Antarctica in 2014 and 2016.**

***Index Terms*—Co-registration, radar sounder, multitemporal analysis, cryosphere, image processing.**

I. Introduction

Radar Sounders (RSs) are active sensors used to explore the subsurface of the cryosphere by generating profile cross-sections, known as radargrams. Existing automatic techniques for automatically analyzing radargrams primarily focus on extracting detailed information about subsurface geology from single-pass RS acquisitions [1] without utilizing the multitemporal data available from repeated acquisitions. This is due mainly to the limited number of multitemporal cryosphere RS campaigns with fully or partially overlapping footprints [2] and the difficulties in co-registering multitemporal radargram pairs. Even when campaigns are planned to overlap, minor variations in latitude and longitude can occur due to the acquisition process and aircraft fluctuations, resulting in radar footprints not entirely overlapping. Multipass RS tracks are intertwined and can fall into three cases (see Fig. 1): i) tracks intersect, RS footprints entirely overlap, and radargrams image a similar subsurface. ii) The tracks are distant, the footprints do not overlap, and the radargrams contain distinct information. iii) The tracks are not far apart, the footprints partially overlap, and the radargrams share similar information. Additionally, the RSs often have different instrument parameters and configurations, which lead to varying geometrical resolutions and geophysical observables [3]. Despite these challenges, multitemporal analyses are crucial for a deeper understanding of cryosphere evolution and monitoring in the context of climate change.

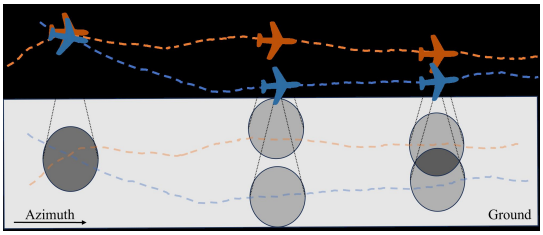


Fig. 1. Acquisition geometry of two tracks and their footprints. From the left, the footprints are fully overlapping, not overlapping, and partially overlapping.

Preliminary studies on multitemporal RS data analysis focused on i) potential improvements in multitemporal data acquisition strategies [2], ii) multitemporal analysis of the percolation zone in accumulation radar data [4], iii) differential interferometry (DInSAR) of multipass RS data [5–7], and iv) analysis of non-coherent multitemporal RS data [8]. In [4], repeated accumulation radar acquisitions of the Greenland firn were analyzed using coherent change detection to identify percolation and refreezing in porous areas. The study emphasized that the key factors for the multitemporal analysis of accumulation data are co-registration and compensation for ice motion. Authors in [6, 7] used DInSAR to analyze coherent RS data, enabling accurate vertical displacement estimation of ice layers while correcting for spatial baseline variations and using precise trajectory data to estimate cross-track layer slopes. However, the interferometric phase is only partially compensated in same-day interferograms, and co-registration errors are not assessed. Finally, [8] presents initial findings on change detection in multitemporal RS data, focusing on the co-registration of non-coherent RS data. Although the results are promising, the phase component is completely discarded. Since the phase contains distinct information from the amplitude, omitting it could lead to greater residual misregistration.

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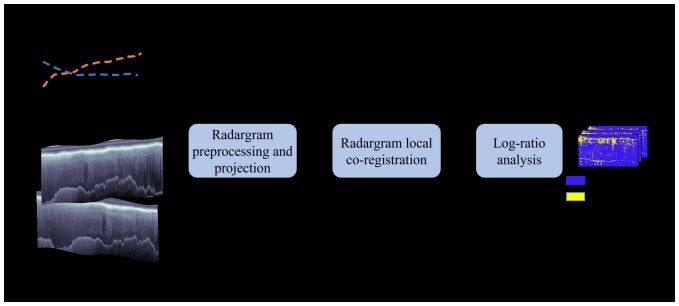


Fig. 2. Flow chart of the co-registration approach.

This study builds upon the work presented in [8] and examines the co-registration accuracy between bi-temporal RS acquisitions with coherent and non-coherent approaches, aiming to evaluate the role of the phase component. Radargrams with full or partial radar footprint overlap are projected and co-registered to mitigate changes due to the ice motion in the along-track direction, vertical variations in the mass balance, and the medium propagation properties. We evaluate the performance by comparing co-registered radargrams using the log-ratio operation. The log ratio highlights the backscattering differences between radargrams related to residual misregistration or morphological changes in the subsurface, if any. The results show high co-registration accuracy for the coherent and non-coherent cases of overlapping radargrams acquired in Antarctica in 2014 and 2016.

II. Methodology

This paper examines the co-registration accuracy between bi-temporal RS data acquired at timestamps t_1 and t_2 with coherent and non-coherent approaches, aiming to evaluate the role of the phase component in the process. We should first mitigate the radargram differences from expected horizontal and vertical changes in the ice pack to identify misregistration. The horizontal displacement is primarily caused by the ice sheet sliding toward the coast. As a result, radargrams acquired at different timestamps, with tracks perpendicular to the sliding direction, may capture different portions of the ice sheet that are still expected to be similar. Radargrams acquired with tracks parallel to the sliding direction image the same portion of the ice sheet but with a forward or backward shift. Horizontal displacement can be mitigated by cross-correlating the radargrams to identify the matching sections that image the same part of the ice sheet. Vertical (range direction) offsets arise from platform fluctuations, differences in radar settings between acquisitions, changes in the ice sheet's mass balance over time [4], and variations in the dielectric properties of the ice sheet. Mass balance changes are related to snow accumulation, flow, and ablation processes over the years and ice flow (e.g., the overburden pressure causing ice divergence or the confluence of two ice streams).

Fig. 2 illustrates the workflow: radargrams are first pre-processed and projected to cover the same subsurface area and have equal length. They are then co-registered using the bedrock as a reference to identify horizontal and vertical offsets related to changes due to horizontal ice motion and vertical variations. Radargrams are locally co-registered using non-coherent and coherent cross-correlation for each trace. Finally, we use the log ratio to estimate the residual misregistration and changes, if any.

A. Radargram pre-processing and projection

Similarly to [8], pre-processing identifies the traces of both radargrams spanning similar latitudes and longitudes. By analyzing the latitude and longitude of the tracks, we determine the common starting and ending coordinates as the first and the last intersection of the tracks and remove the other traces. Additionally, we calculate the average track distance and remove traces farther apart than the antenna pulse-limited footprint to ensure that the radargrams image the same subsurface area. The result is two radargrams acquired over a similar area but with differing numbers of columns.

Each track is defined by the latitude and longitude sequences $X_{t_i} = \{x(a) \mid a \in [1, N_T^i]\}$ and $Y_{t_i} = \{y(a) \mid a \in [1, N_T^i]\}$, where N_T^i is the number of radargram traces (columns). Similarly to [8], we project the longest radargram (slave) onto the shortest one (master), applying dynamic time warping (DTW) [9]. Let us suppose without loss of generality that the track in t_1 is longer than that in t_2—thus, the track in t_1 is the slave, and the track in t_2 is the master. DTW analyses the slave $X_{t_1}^s, Y_{t_1}^s$ and master X_{t_2}, Y_{t_2} and performs the optimal matching with the following rules– i) every value in $X_{t_1}^s, Y_{t_1}^s$ must be matched with those of X_{t_2}, Y_{t_2}. ii) The first value of $X_{t_1}^s, Y_{t_1}^s$ must be matched with the first value of X_{t_2}, Y_{t_2}. iii) The last value of $X_{t_1}^s, Y_{t_1}^s$ must be matched with the last value of (X_{t_2}, Y_{t_2}). iv) The mapping of $X_{t_1}^s, Y_{t_1}^s$ to X_{t_2}, Y_{t_2} must be monotonically increasing– given two coordinates $x_{t_1}, x'_{t_1} \in X_{t_1} \mid x_{t_1} > x'_{t_1}$ and $y_{t_1}, y'_{t_1} \in Y_{t_1} \mid y_{t_1} > y'_{t_1}$ then there must not be two values $x_{t_2}, x'_{t_2} \in X_{t_2} \mid x_{t_2} > x'_{t_2}$ and $y_{t_2}, y'_{t_2} \in Y_{t_2} \mid y_{t_2} > y'_{t_2}$, such that x'_{t_1} and y'_{t_1} are matched with x_{t_2} and y_{t_2} respectively, and x_{t_1} and y_{t_1} are matched with x'_{t_2} and y'_{t_2} respectively. The optimal match satisfies all criteria and minimizes the cost. It is calculated as the sum of absolute distance for each pair of matched latitude and longitude values. After the optimal match, the slave radargram and latitude and longitude sequences are projected accordingly. The results are two radargrams with the same number of traces.

B. Co-registration

As output of the projection step, we obtain two coherent radargrams $\mathbf{R}_{t_1}$ and $\mathbf{R}_{t_2}$ with fully or partially overlapping footprints. Each radargram is a 2D matrix of N_T^i traces and N_S^i samples (rows):

$$\mathbf{R}_i = \{R_i(r, a) \mid a \in [1, N_T^i], r \in [1, N_S^i]\}, \tag{1}$$

where $i = t_1, t_2$, R_i is the coherent signal from the surface and subsurface, and (a, r) are the azimuth and range coordinates, respectively. Each trace $a \in [1, N_T^i]$ in the radargrams is specified in a longitude/latitude reference system by latitude $x(a) \in (-90, +90]$ and longitude $y(a) \in (-180, +180]$.

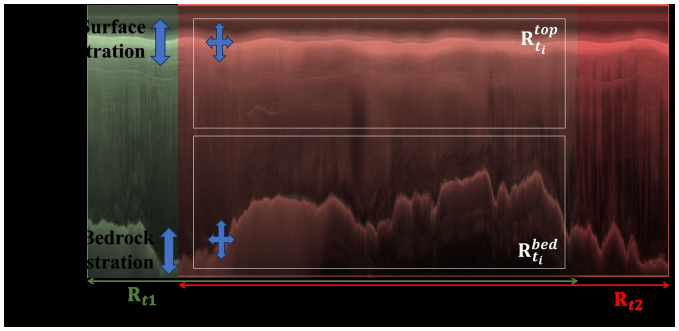


Fig. 3. Cross-correlation geometry of radargrams $\mathbf{R}_{t_1}$ in green and $\mathbf{R}_{t_2}$ in red– the white boxes indicate the sub-radargrams, and blue arrows indicate the cross-correlation directions.

Projected radargrams may exhibit expected changes due to horizontal and vertical displacements of the ice sheet and platform fluctuations [4]. These displacements must be estimated before analyzing the misregistration. To estimate the horizontal and vertical displacements, we perform a first coarse co-registration using the bedrock as a reference (assuming its stability in t_1 and t_2) that resolves constant horizontal and vertical offsets among the radargrams. We then perform a second co-registration focusing on the local areas of the radargrams. This is because the vertical dimensions of radargrams $\mathbf{R}_{t_1}(r,a)$ and $\mathbf{R}_{t_2}(r,a)$ depend on the dielectric properties and thermal regime of the ice sheet in t_1 and t_2 and the acquisition geometry. Additionally, the sliding velocity of the ice varies with the depth. Hence, we horizontally divide the radargrams into two sub-radargrams $\mathbf{R}_{t_i}^{\text{top}}(r,a)$ and $\mathbf{R}_{t_i}^{\text{bed}}(r,a)$ that are separately analyzed (see Fig. 3). The upper one $\mathbf{R}_{t_i}^{\text{top}}(r,a)$ images the surface and the firn and part of the ice stratigraphy. The lower sub-radargram $\mathbf{R}_{t_i}^{\text{bed}}(r,a)$ images the basal area and the basal interface.

$\mathbf{R}_{t_i}^{\text{top}}(r,a)$ and $\mathbf{R}_{t_i}^{\text{bed}}(r,a)$ are separately co-registered to estimate the vertical offset for each trace with a coherent and non-coherent approach to investigate the importance of the phase. Both sub-radargrams $\mathbf{R}_{t_i}^{j}(r,a), j = \text{top, bed}$ are coherently co-registed considering the 2D cross-correlation:

$$\gamma(r',a') = \sum_{r,a} \mathbf{R}_{t_1}^{j}(r,a)\,\overline{\mathbf{R}}_{t_2}^{j}(r-r',a-a') \tag{2}$$

where $\overline{\mathbf{R}}_{t_2}^{j}$ denotes complex conjugation of $\mathbf{R}_{t_2}^{j}$, and $j = \text{top, bed}$. The formula for the non-coherent approach can be simplified similarly to the work in [8]. Before the cross-correlation, radargrams are separately normalized by scaling the magnitude while maintaining the phase to simplify the calculation. To co-register the sub-radargrams, we calculate the offset of each trace that maximizes the cross-correlation coefficient γ. The offsets are then applied to the slave radargram to obtain the co-registered one $\hat{\mathbf{R}}_{t_1}^{j}(r,a)$.

C. Co-registration validation

Once that both sub-radargrams radargram $\hat{\mathbf{R}}_{t_1}^{j}(r,a)$ and $\mathbf{R}_{t_2}^{j}(r,a)$ are co-registered with the coherent and non-coherent approach, we estimate the residual misregistration as log ratio:

$$\mathcal{L}^{j}(a,r) = \log \frac{|\hat{\mathbf{R}}_{t_1}^{j}(r,a)|}{|\mathbf{R}_{t_2}^{j}(r,a)|}. \tag{3}$$

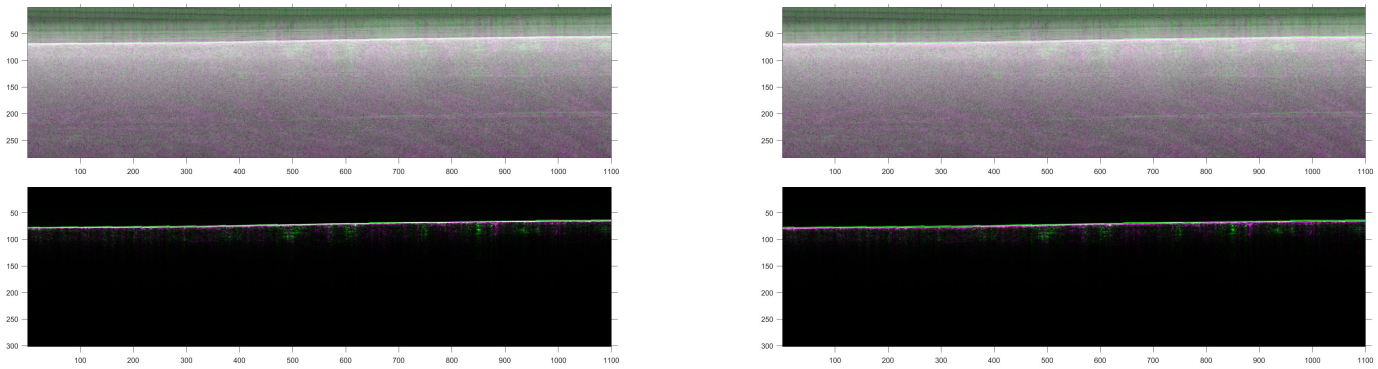
The log ratio is a well-known measure in the SAR literature and highlights the differences in the radar images while reducing the impact of noise (speckle) and other artifacts. The log-ratio image $\mathcal{L}^{j}(a,r)$ is expected to have small values in areas with perfect co-registration and no changes and high values in areas with residual misregistration or affected by change, if any. Hence, we thresholded $\mathcal{L}^{j}(a,r)$ with Otsu's method [10] on a per-pixel basis to identify misregistered areas or possible changes in the ice pack.

III. Dataset description and Experimental results

We analyze two campaigns with overlapping antenna footprints in Antarctica by the MCoRDS-3 instrument [11] from the Center for Remote and Sensing and Integrated Systems (CReSIS). Here, for space constraints, we show part of radargram frame 14 of campaigns 20141103-04 and 20161103-06, acquired in 2014 and 2016, respectively. The data are pre-processed with range compression, azimuth focusing with synthetic aperture radar (SAR), and array processed using delay-and-sum beamforming with all RS channels [11]. To quantitatively validate the coherent and non-coherent co-registration, we consider the normalized cross-correlation coefficient (NCC), normalized mutual information (NMI), and the structural similarity index (SSIM) as metrics. The metrics range in $[0,1]$, and a higher value indicates a more significant similarity between the radargrams.

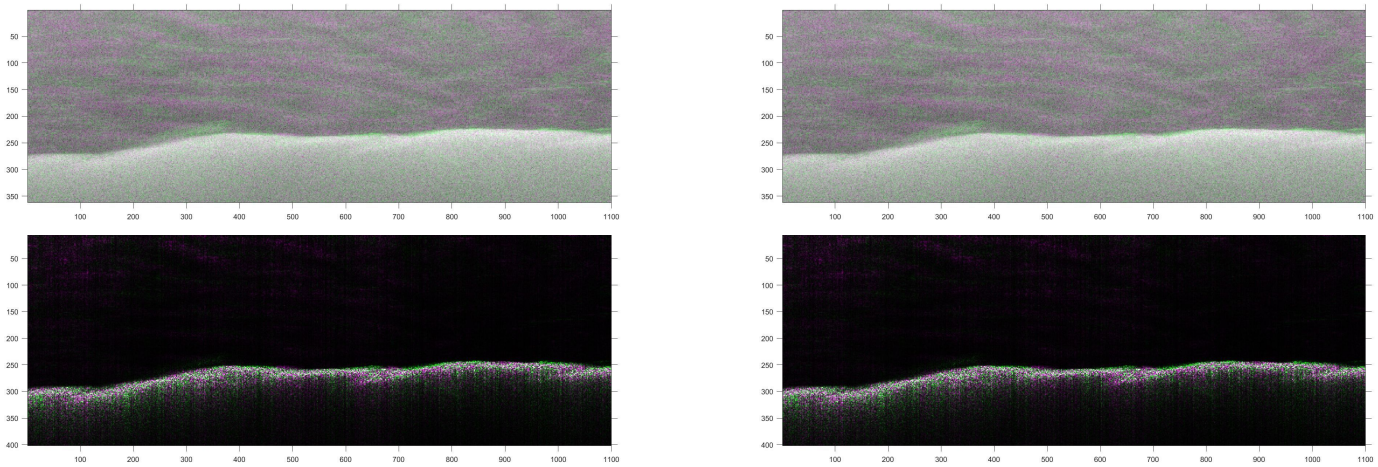
The qualitative results in Fig. 4 and 5 show in the first row the multitemporal false color composition of the radargrams (R - 2014, G - 2016, B - 2014) using the non-coherent (left) and the coherent (right) approaches. The second row shows the thresholded log ratio of the two co-registered radargrams. We highlight in green and magenta the variation of backscattering values in 2014 and 2016, respectively. In both sub-radargrams, the non-coherent and coherent approaches accurately correlate the radargrams with minimal residual misregistration in the strongest reflections, i.e., the surface and the bedrock. Moreover, both the multitemporal false color composition and the log-ratio images show discrepancies between the 2014 and 2016 radargrams that cannot be associated with misregistration and may be generated by changes in the ice pack.Coherent results show slightly better co-registration and backscattering differences (including misregistration) in the log-ratio image.

The quantitative results are reported in Table I for both sub-radargrams that are part of radargram frame 14 co-registered with the non-coherent and coherent approach. Note that we masked the area above the surface to calculate the metrics for the top ice sub-radargrams, as the signal mostly depends on the transmitted chirp and not on ice reflections. The quantitative results confirm the goodness of the co-registration based on the non-coherent and coherent approaches. However, they showed slightly better performance with the coherent approach.



(a) Non-coherent approach. (b) Coherent approach.

Fig. 4. Experimental results for $\mathbf{R}_{t_2}^{\text{top}}(r, a)$ for the non-coherent (a) and the coherent (b) approaches. Images in the top row are the multitemporal false color compositions of co-registered radargrams (R - 2014, G - 2016, B - 2014). Images in the bottom line are the log ratios, where the green and magenta highlight the variations of backscattering values in 2014 and 2016, respectively.



(a) Non-coherent approach. (b) Coherent approach.

Fig. 5. Experimental results for $\mathbf{R}_{t_2}^{\text{bed}}(r, a)$ for the non-coherent (a) and the coherent (b) approaches. Images in the top row are the multitemporal false color compositions of co-registered radargrams (R - 2014, G - 2016, B - 2014). Images in the bottom line are the log ratios, where the green and magenta highlight the variations of backscattering values in 2014 and 2016, respectively.

TABLE I
QUANTITATIVE ANALYSIS OF THE CO-REGISTRATION WITH A NON-COHERENT (NC) AND COHERENT (C) FUNCTION.

Metrics	Non-Coherent		Coherent	
	$\mathbf{R}_{t_2}^{\text{top}}$	$\mathbf{R}_{t_2}^{\text{bed}}$	$\mathbf{R}_{t_2}^{\text{top}}$	$\mathbf{R}_{t_2}^{\text{bed}}$
NCC	0.7360	0.8300	0.7600	0.8799
NMI	0.8935	0.7749	0.9149	0.7749
SSIM	0.9886	0.9971	0.9882	0.9972

IV. CONCLUSIONS AND FUTURE WORKS

This paper examines the co-registration accuracy in bi-temporal remote sensing acquisitions, using coherent and non-coherent approaches to assess the impact of the phase component. We project and co-register radargram pairs with overlapping footprints to mitigate the horizontal changes related to the ice motion and vertical variations in mass balance. Additionally, we use the log ratio to highlight residual misregistration or morphological changes, if any.

We analyzed radargrams acquired in Antarctica and spanning the same track in 2014 and 2016. The results show that the misregistration for the coherent and non-coherent cases is limited and mainly focuses on the areas with bright reflections, such as the surface and the base interface. Finally, the coherent co-registration performs slightly better.

In future work, we aim to perform sub-pixel registration to achieve finer co-registration, which can be done by oversampling or peak-finding the cross-correlation matrix. Additionally, we want to apply the co-registration to a large number of data and design novel methods to monitor the changes in the ice pack.

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